Physics 5450: Quiz #01

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Calculate the probability P_n to find a particle, which is in the n-th stationary state of the infinite square well (extending from 0 to a), between $x_1 = a/4$ and $x_2 = a/3$.

(a) Try to get the answer for general n. If this is too difficult, calculate P_n for $n = 1, 2, 3$. [15 pts.]

(b) Sketch first three eigenstates $\psi_n(x)$ and argue in favor of the result (a). [5 pts.]

(a) $$P_n = \int_{a/4}^{a/3} dx\, |\psi_n|^2 = \int_{a/4}^{a/3} dx\, \frac{2}{a} \sin^2\left(\frac{\pi n x}{a}\right) = \frac{2}{a}\,\frac{a}{\pi n} \int_{\pi n/4}^{\pi n/3} dy\, \sin^2 y$$

$$y = \frac{\pi n x}{a}$$

Use $\sin^2 y = \frac{1}{2}(1 - \cos 2y)$

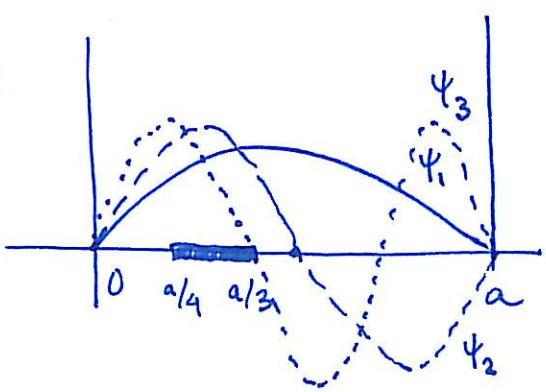
$$P_n = \frac{1}{\pi n} \int_{\pi n/4}^{\pi n/3} dy\, (1 - \cos 2y) = \frac{1}{\pi n}\left(y - \frac{1}{2}\sin 2y\right)\Big|_{\pi n/4}^{\pi n/3} =$$

$$= \frac{1}{\pi n}\left(\frac{\pi n}{3} - \frac{\pi n}{4} - \frac{1}{2}\left(\sin\frac{2\pi n}{3} - \sin\frac{2\pi n}{4}\right)\right) =$$

$$= \frac{1}{12} - \frac{1}{2\pi n}\left(\sin\frac{2\pi n}{3} - \sin\frac{\pi n}{2}\right).$$

$P_1 = 0.1$, $P_2 = 0.15$, $P_3 = 0.03$

(b)



Probability in question is given by the area under $|\psi_n|^2$ curve. From the figure, it is clear that it should be maximal for $n=2$ and minimal for $n=3$. $|\psi_1|^2$ is too small in that interval as it reaches its maximum at the midpoint $a/2$.